

The **optional** qualifier MAY be used to declare a field which can be omitted from the structure encoding under some circumstances.

B.3.9.2. CHOICE OF Fields

A field within a **STRUCTURE** definition MAY be defined to be a **CHOICE OF** (either directly within the **STRUCTURE** definition or via a type reference). Over the wire, such a field is encoded as one of the alternate types given in the **CHOICE OF** definition. For example, the **user-id** field in the following **STRUCTURE** MAY be encoded as either a TLV **UTF-8 String** or an **Unsigned Integer**.

STRUCTURE with CHOICE OF Field Example

```
user-information => STRUCTURE [ extensible ]
{
  user-id [1] : CHOICE OF
  {
    UNSIGNED INTEGER,
    STRING
  }
}
```

If a **tag** qualifier is given for a **CHOICE OF** field (e.g. **[1]** as shown above), that tag SHALL be used in the encoding of the field for all possible alternates. On the other hand, if a **tag** qualifier is *not* given, then the default tag associated with the selected **CHOICE OF** alternate SHALL be used in the encoding. For example, in the following structure, a context-tag of **1** will be encoded if the **user-id** field is an Unsigned Integer, or **2** if the field is a String.

STRUCTURE with CHOICE OF Field with Default Tag Example

```
user-information => STRUCTURE [ extensible ]
{
  user-id : CHOICE OF
  {
    id [1] : UNSIGNED INTEGER,
    name [2] : STRING
  }
}
```

Note that, in all cases, the tag or tags associated with a **CHOICE OF** field SHALL be unique within the context of the containing **STRUCTURE**.

B.3.9.3. Includes FIELD GROUP

A **includes** statement MAY be used within a **STRUCTURE** definition to incorporate the fields of a **FIELD GROUP** defined outside the **STRUCTURE**. The fields of the **FIELD GROUP** are included in the **STRUCTURE** as if they had been listed within the **STRUCTURE** definition itself.